

CET 479
Spring 2023
In class Activity – Feb 22, 2023.

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LAB_05 : Use commands in Linux environment

1. Show the utilization of the following commands using your Linux environment.

- a. **mv** : Renames or Moves one or more files.

There are three forms:

- The first : Renames a single file with a new filename that you supply.
- The second : Renames one or more files so that they appear in a specified directory.
- The third : Renames a directory.

- `$[mv <file name ><path name >]` : move a **file** to new place.
- `$[mv <old name ><new name >]` : can use mv for change name of file.
- `$[mv -f <file name ><path>]` : (force) not to prompt if would overwrite file.

`$[mv -i <file name ><path>]` : (interactive) prompts for confirm overwrite a file.

`$[mv -v <file name ><path>]` : (verbose) lists files as they are moved.

```
tmpinclass
├── EXCISE_01
│   ├── excise_01_1
│   ├── excise_01_2
│   └── excise_01_3
├── EXCISE_02
│   ├── dir_file_list.txt
│   ├── dir_listing.txt
│   ├── excise_02_1
│   ├── excise_02_2
│   └── excise_02_3
├── excise_02_1
├── excise_02_2
├── excise_02_3
└── EXCISE_03

3 directories, 11 files
[van@localhost ~]$ mv tmpinclass/excise_02 tmpinclass/EXCISE_02/
[van@localhost ~]$ tree tmpinclass
tmpinclass
├── EXCISE_01
│   ├── excise_01_1
│   ├── excise_01_2
│   └── excise_01_3
├── EXCISE_02
│   ├── dir_file_list.txt
│   ├── dir_listing.txt
│   ├── excise_02_1
│   ├── excise_02_2
│   └── excise_02_3
├── excise_02_1
├── excise_02_2
├── excise_02_3
└── EXCISE_03

3 directories, 11 files
[van@localhost ~]$ _
```

b. grep

- `$[grep <file name>]:` Searches for a Pattern in Files
- `$[grep -c <file name>]:` (count) displays the number of lines that contain in each file.
- `$[grep -color <file name>]:` to display the match in color on a terminal.
- `$[grep -E <file name>]:` (extended) to accept extended regular expressions.
- `$[grep -i <file name>]:` (-ignore-case) causes lowercase letters in the pattern to match uppercase letters in the file, and vice versa.
- `$[grep -l <file name>]:` (lowercase) A filename is displayed only once.
- `$[grep -R <file name>]:` (recursive) searched and processes files within these directories.
- `$[grep -v <file name>]:` (Invert-match) displays all lines that do not contain a match for the pattern
- `$[grep -w <file name>]:` (word-regexp) must match a whole word.

```
3 directories, 11 files
ivan@localhost ~1$ mv -i tmpinclass/EXCISE_02/excise_02_1 tmpinclass/EXCISE_03/change_name.txt
mv: cannot stat 'tmpinclass/EXCISE_02/excise_02_1': No such file or directory
ivan@localhost ~1$
ivan@localhost ~1$ grep tmpinclass/EXCISE_02/excise_02_02
^Z
[1]+  Stopped                  grep --color=auto tmpinclass/EXCISE_02/excise_02_02
ivan@localhost ~1$
```



c. **cp** : Copies and rename one or more files

- **\$[cp -i]** : (*interactive*) option prompts you before it overwrites a file.
- **\$[cp -a]** : (*archive*) option attempts to preserve the owner
- **\$[cp -b]** : (*backup*) option makes a backup copy of a file.
- **\$[cp -f]** : (*force*) option to remove the destination file.
- **\$[cp -p]** : (*preserve[=attr]*) creates a destination file with the same owner.
- **\$[cp -r]** : (*recursive*) copies directory hierarchies.
- **\$[cp -v]** : (*verbose*) option displays the name of each file as copies it.

```

3 directories, 11 files
van@localhost ~]$ mv -i tmpinclass/EXCISE_02/excise_02_1 tmpinclass/EXCISE_03/change_name.txt
mv: cannot stat 'tmpinclass/EXCISE_02/excise_02_1': No such file or directory
van@localhost ~]$ 
van@localhost ~]$ grep tmpinclass/EXCISE_02/excise_02_02
^Z
11+ Stopped                  grep --color=auto tmpinclass/EXCISE_02/excise_02_02
van@localhost ~]$ cp tmpinclass/EXCISE_02/excise_02_? tmpinclass/EXCISE_03
van@localhost ~]$ tree tmpinclass
tmpinclass
├── EXCISE_01
│   ├── excise_01_1
│   ├── excise_01_2
│   └── excise_01_3
├── EXCISE_02
│   ├── dir_file_list.txt
│   ├── dir_listing.txt
│   ├── excise_02_2
│   ├── excise_02_3
│   ├── excise_03_1
│   ├── excise_03_2
│   └── excise_03_3
└── EXCISE_03
    ├── change_name.txt
    ├── excise_02_2
    └── excise_02_3

3 directories, 13 files
van@localhost ~]$

```



d. **Ln** : Creating a hard or soft link for files

- `$[ln <existing file> <link hard file> ]` : create a link hard file.
`$[ls -li <linking file>]` : Verify linking file with inode number.
- `$[ln -s <existing file> <link soft file> ]` : create a link soft file
- `$[ln -sf <link soft file> ]` : to create a symbolic link overwrites the destination file or directory if it already exists, without prompting the user for confirmation
- `$[ln -sb <existing file> <link soft file> ]` : create a link soft file with back up.

```

3 directories, 13 files
[van@localhost ~]$ cd tmpinclass
[van@localhost tmpinclass]$ cd EXCISE_01
[van@localhost EXCISE_01]$ ln excise_01_1 Link_hard_01
[van@localhost EXCISE_01]$ ls -li
16920397 excise_01_1 16920399 excise_01_2 16920401 excise_01_3 16920397 Link_hard_01
[van@localhost EXCISE_01]$ ln -s excise_01_1 Link_sodt_02
[van@localhost EXCISE_01]$ ls -li
16920397 excise_01_1 16920401 excise_01_3 16920404 Link_sodt_02
16920399 excise_01_2 16920397 Link_hard_01
[van@localhost EXCISE_01]$ ln -sf excise_01_1 Link_soft_02
ln: invalid option -- '-'
Try 'ln --help' for more information.
[van@localhost EXCISE_01]$ ls -li
16920397 excise_01_1 16920401 excise_01_3 16920404 Link_sodt_02
16920399 excise_01_2 16920397 Link_hard_01
[van@localhost EXCISE_01]$ ln -sf excise_01_1 Link_soft_02
[van@localhost EXCISE_01]$ ls -li
16920397 excise_01_1 16920401 excise_01_3 16920404 Link_sodt_02
16920399 excise_01_2 16920397 Link_hard_01 16920405 Link_soft_02
[van@localhost EXCISE_01]$ _

```



e. **file**

- `$(file <file name>)` : Displays the Classification of a File.
- `$(file -i <file name>)` : (mime) displays type strings.
- `$(file -z <file name>)` : (Zip) to classify files within a compressed file.

```
3 directories, 16 files
[van@localhost ~]$ file tmpinclass
tmpinclass: directory
[van@localhost ~]$ file tmpinclass/EXCISE_01/excise_01_1
tmpinclass/EXCISE_01/excise_01_1: empty
[van@localhost ~]$ file -i tmpinclass/EXCISE_01/excise_01_1
tmpinclass/EXCISE_01/excise_01_1: inode/x-empty; charset=binary
[van@localhost ~]$ file -z tmpinclass/EXCISE_01/excise_01_1
tmpinclass/EXCISE_01/excise_01_1: empty
[van@localhost ~]$
```

f. **Pipe operator** : Send the output of one command to the input of another

- `$(touch <file name> | ls)`

```
3 directories, 16 files
[van@localhost ~]$ file tmpinclass
tmpinclass: directory
[van@localhost ~]$ file tmpinclass/EXCISE_01/excise_01_1
tmpinclass/EXCISE_01/excise_01_1: empty
[van@localhost ~]$ file -i tmpinclass/EXCISE_01/excise_01_1
tmpinclass/EXCISE_01/excise_01_1: inode/x-empty; charset=binary
[van@localhost ~]$ file -z tmpinclass/EXCISE_01/excise_01_1
tmpinclass/EXCISE_01/excise_01_1: empty
[van@localhost ~]$ cd EXCISE_03
-bash: cd: EXCISE_03: No such file or directory
[van@localhost ~]$ cd tmpinclass/EXCISE_03
[van@localhost EXCISE_03]$ touch excise_02_2 | ls
change_name.txt excise_02_2 excise_02_3
[van@localhost EXCISE_03]$ _
```



g. `finger` and `w` (see if these work with the new version of fedora we are currently using)

`finger`

- `{ finger | f } [[-l] [-p]] [[-i] [-q] [-s] [-w]]`
- `$[finger @<host name>]` : To get information all users logged in to host
- `$[finger <name>@<host name>]` : `name` user logged in to host

- **W** : Allows administrators to view information about users that are currently logged in.

`$[ w <options><users name>]`

Options:

- `-h` : print output no header
- `-u` : Ignores user name
- `-s` : Short format
- `-f` : from (remote host name) help
- `-i` : Replace the hostname in the FROM field with the IP address
- `-v` : Version.
- `-o` : Old-style out put

```
phoenixnap@test-system:~$ w -s
08:48:42 up 2:19, 1 user, load average: 0.14, 0.03, 0.01
USER      TTY      FROM          IDLE WHAT
phoenixn  :0        :0             ?xdm? /usr/lib/gdm3/gdm-x-session --run-script env GN
phoenixnap@test-system:~$
```

```
phoenixnap@test-system:~$ w -h
phoenixn  :0        :0             06:30  ?xdm? 1:30 0.00s /usr/lib/gdm3/gdm-x-session
phoenixnap@test-system:~$
```

h. **Locate** : (Finds files) is used for obtaining instantaneous results, and it is an essential utility when speed is a priority.

- `$[locate <file name>]` : Find a file .
- `$[locate <file name> | less]` : Format the Output
- `$[locate -e <file name>]` : to search the filesystem and include only the existing files in the results.
- `$[locate -c <file name>]` : Count the Existing Files.
- `$[locate -d <file name>]` : Disable Errors While Searching.
- `$[locate -n <file name> <number>]` : Limit the Number of Search Results.
- `$[locate -i <file name>]` : search ignores case-sensitivity.
- `$[locate -r <file name>]` : Search for a File with an Exact Name.
- `$[locate -o <file name>]` : Separate Output .
- `$[locate -s <file name>]` : View Database Statistics.

2. Perform the following tasks.

a. Redirecting output to a text file (> operator) and “ | ”

`$[ls -al > <file name>]` : re-directed to file, instead of the screen

`$[cat <file name>]` : verify for re-directed to file

```
home@VirtualBox:~$ ls -al > listings
home@VirtualBox:~$ cat listings
total 324
drwxr-xr-x 26 home home 4096 2012-09-10 10:42 .
drwxr-xr-x  3 root root 4096 2012-09-01 19:43 ..
-rw-rw-r--  1 home home    0 2012-09-10 09:25 abc
```

\$[**practice homework cet_479 >>** <file name>] : Redirecting some word (output) to a text file

```
home@VirtualBox:~$ cat sample
Hang on for the best Linux Lessons.
home@VirtualBox:~$ echo Thanks for reading >> sample
home@VirtualBox:~$ cat sample
Hang on for the best Linux Lessons.
Thanks for reading
```

\$ [**cat music.mp3 >** /dev/audio] : output to a mp3 file

\$ **ls -la | less** : To display output of a directory listing one page at a time

b. Redirecting input from a text file (< operator)

\$ **mails -s "Subject" van@gmail.com <** <file name> : Sent a file attachment to address email.

Attachment File

```
guru99@VirtualBox:~$ mail -s "News Today" abc@ymail.com < NewsFlash
```

E-mail Subject E-mail Address

c. Search for two different words in a file (write the command and also display how to use it.)

\$ [**grep -wi 'word1\|word2'** <file name>] > <file name out put>

